

Shivnarayan Dhuppar

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RESEARCH INTEREST

Immunology, Cancer, Gene Expression, Computational Methods, Microscopy

EDUCATION

- 2013 - 2020** **Tata Institute of Fundamental Research, India**
PhD, Physics
Thesis: On Cell Cycle Dependent DNA Damage Responses and Gene Expression as Assayed from Microscopic Image Analysis
- 2010 - 2013** **The Maharaja Sayajirao University of Baroda, India**
B.Sc. (Hons) Physics
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RESEARCH EXPERIENCE

- 2023 - present** **Harvard Medical School and Massachusetts General Brigham**
Long-term health effects of prenatal stress (MGH)
Exposure to prenatal stress interferes with brain development. I am investigating the pathways that can be targeted to reduce if not reverse the adverse effects of prenatal stress using single-cell sequencing technologies and brain interventions (stereotaxic surgeries). The project is part of the Biology of Adversity Project at the Broad Institute of MIT and Harvard.
- Roles of miR-92a in colitis and colorectal cancer (BWH)**
miR-92a from the oncogenic miR 17-92 cluster is upregulated in many cancers, including colorectal cancer. In this project, I investigated the roles of miR-92a in colitis and colorectal cancer using miR-92a KO mice models. The project formed the basis of an (now successful) R01 application for the PI.
- Roles of miR-21 in Multiple Sclerosis (BWH)**
In this collaborative project I investigated how miR-21, a pro-inflammatory microRNA often associated with immune activation/autoimmunity, drives pathogenic Th17 cells, showing that Type I interferon limits CNS autoimmunity by regulating the miR-21–FOXO1 axis.
- 2020 - 2021** **Children's Hospital of Philadelphia, USA**
At CHOP, I spent 6 months post PhD to extend my thesis work on the cell cycle-dependent changes in genome architecture using chromosome capture technology and Crispr-knock-ins.
- 2014 – 2019** **Tata Institute of Fundamental Research (TIFR), India**
Cell Cycle Staging from Image Analysis
In this project I developed a fully automated image analysis routine to stage cells in the cell cycle from high-resolution microscopy images. The module was used to study cell cycle-dependent nuclear architecture, and DNA damage response at single cell resolution by combining DNA FISH with RNA FISH and IF (resulting in two corresponding author papers).
- Studying Defects in Crystalline Media Using Algebraic Topology**
In this project, I modeled ordered, periodic systems such as liquid crystals in terms of their underlying molecular symmetries, which define the appropriate order parameters. This formalism enabled a rigorous study of defects—singularities in the order-parameter space—using tools from algebraic topology.
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AWARDS, HONORS AND FELLOWSHIPS

- 2021** **Indian Institute of Science**
Institute of Eminence Postdoctoral Fellowship (declined)
- 2020** **Department of Science and Technology, Government. of India**
AWSAR Award for best popular science article

- 2019 Science Engineering and Research Board, India**
Travel grant for presenting our study at Single Cell Analyses conference hosted by Cold Spring Harbor Laboratory
- 2015 Department of Atomic Energy, Government of India**
Senior Research Fellowship for years 2015 - 2019
- 2013 Department of Atomic Energy, Government of India**
Junior Research Fellowship for years 2013 - 2015
- 2012 Circuits and Electronics (6.002x) from MITx**
Part of the founding batch with Prof. Anant Agarwal.
- 2011 A R Rao Mathematical Society**
First in the State Math Olympiad
- 2010 National Board for Higher Mathematics**
Award for distinctive performance at Mathematics Training and Talent Search

TALKS AND PRESENTATIONS

- 2026 Biology of Adversity Project @ the Broad Institute of MIT and Harvard**
Talk: *Prenatal Stress: Mitigating Long-Term Health Risks—Preclinical Studies, observations, and interventions*
- 2024 Proteomics @ the Broad Institute of MIT and Harvard**
Talk: *Quantitative Microscopy for Capturing Biological Insights*
- 2019 Single Cell Analyses, Cold Spring Harbor Laboratory Meetings**
Talk and poster: *Investigating Central Dogma in the Context of Nuclear Architecture and Cell Cycle at Single-Cell Resolution*
*Travel supported by Science and Engineering Research Board (SERB), India.
- 2016 Optics Within Life Sciences (OWLS)**
Poster: *Exploring Regulatory Roles of Genomic Organization in Gene Expression and DNA Repair Using Fluorescence Microscopy*
- 2011 Mathematics Training and Talent Search**
Talk: *Geometric Interpretations of Algebraic Operations in the Complex Plane.*

PUBLICATIONS

- 2025 Type I interferon limits central nervous system autoimmunity by modulating the microRNA-21/FOXO1 axis in pathogenic T helper 17 cells**
J. Varghese, A. Cannon, ..., **S. Dhuppar**, ..., M. Gopal, *Science Transl Med*, 2025
- 2025 miRNAs in the Biology and Hallmarks of Neurodegeneration**
S. Dhuppar*, W. Poller, M. Gopal, *Trends in Molecular Medicine*, 2025
- 2024 Combined 3D DNA FISH, Single Molecule RNA FISH and Immunofluorescence**
S. Sen, **S. Dhuppar**, A. Mazumder, *MIMB (Springer) Book Chapter*, 2023
- 2024 Type I IFN signaling limits CNS autoimmunity by modulating miR-21-Foxo1 axis in pathogenic Th17 cells**
M. Fujiwara, J. Varghese, A. Cannon, L. Garo, A. K. Ajay, R. Kadowaki-Saga, G. Gabriely, **S. Dhuppar**, et. al. (in preparation)
- 2023 Protocol for analyzing transforming growth factor β signaling in dextran-sulfate-sodium-induced colitic mice using flow cytometry and western blotting**
M. Fujiwara, L. Garo, A. k. Ajay, A. Cannon, P. Kolyapetri, **S. Dhuppar**, G. Murugaiyan, *Star Protocol*, 2023
- 2022 MicroRNA effects on Gut Homeostasis: therapeutic Implications for Inflammatory Bowel Disease**
S. Dhuppar and M. Gopal, *Trends in Immunology*, 2022

- 2020 *Investigating Cell Cycle-Dependent Gene Expression in the Context of Nuclear Architecture at a Single-Allele Resolution*
S. Dhuppar* and A. Mazumder*, *Journal of Cell Science*, 2020
- 2020 *γ H2AX Peak in the S Phase After UV Irradiation Corresponds to DNA Replication and Does not Report on the Extent of DNA Damage*
S. Dhuppar*, S. Roy and A. Mazumder*, *Molecular and Cellular Biology*, 2020
- 2020 *CTCF Mediated Genome Architecture Regulates the Dosage of Mitotically Stable monoallelic Expression of Autosomal Genes*
 K. R. Chandradoss, B. Chawla, **S. Dhuppar**, R. Nayak, R. Ramachandran, S. Kurukuti, A. Mazumder, and K. S. Sandhu, *Cell Reports*, 2020
- 2018 *Measuring Cell Cycle-Dependent DNA Damage Responses and p53 Regulation on a Cell-by-Cell Basis from Image Analysis*
S. Dhuppar and A. Mazumder, *Cell Cycle*, 2018
- * Corresponding author**

PROFESSIONAL AFFILIATIONS / SERVICES

- 2024 - present **Biology of Adversity Project and the Broad Trauma Initiative**
- 2023 - present **AACR, NYAS, AAAS**
- 2018 - present **Reviewer for Nature, PLOS, Frontiers, Taylor and Francis**
- 2021 - 2023 **4D Nucleome Project, NIH common funds**

KEY SKILLS

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| Programming and data | - High-throughput sequencing data (scRNA seq, ATAC-seq, Hi-C) analysis in R and Python
- SLURM, Google GCP, Apptainer (Singularity), Bash/Linux, C
- AI/ML libraries: Scikit-learn, Gensim, RDKit, PyTorch
- Computer vision (Matlab, Python) |
| Microscopy | - light-sheet microscopy (LaVision UMP II), confocal, FCS, FRET, FRAP, live cell imaging, |
| Mol Biology | - CRISPR knock-ins, siRNA knockdowns, Cloning,
- qPCR, WB, high dimensional flowcytometry (ARIAs, Bigfoot), ELISA
- <i>in situ</i> Hybridization (ISH)-based techniques: smFISH, DNA FISH |
| Animal and cell Models | - Mouse handling, stereotaxic surgeries, cell culture (stem cells, differentiated cells) |
| Theory | - Statistical Physics: theory and simulations (Monte Carlo, Markov) |